INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

Data centres over-cool, continuously, because nobody can promise them the hours ahead. A chiller plant needs hours of notice to change mode, and a thermometer only ever reports NOW -- so the mechanical chillers keep running through hours that outside air could have cooled for free. FortyGuard closes that gap with heat intelligence 2 m above the ground, the height a ground-mounted condenser actually breathes. This agent turns that forecast into an hour-by-hour schedule carrying a calibrated safety bound.

THE SITE

Location	Amazon Web Services, OR -- station KPSC, America/Los_Angeles
Committed pair	OSM 1252196821 -> 1252196820 Amazon Web Services / Amazon Web Services
Facade gap	73.7 m between the two halls
Weather record	43,732 real hourly records from KPSC
Plume physics	576 steady-state solves on this site's own footprints; worst intake rise 0.3411 C at 90 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day	2022-06-11 (recirc_edge)
Plant limit	21.0 C
Notice required	3 h
Forecast skill	0.50 relative to persistence
Level anchor	sensor
Condenser bank	longest facade
Switch budget	2 mode changes per day
Minimum dwell	3 h
Max dew point	15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 1 of 24 hours with 1 mode change. That is 1 more than the reactive incumbent, at 0 unsafe hours against its 0. 2 hours were safe but still ran chillers, because the schedule could not afford them.

Agent	1 of 24 hours free cooling, 1 mode change(s), 0 unsafe hour(s)
Incumbent	0 of 24 hours free cooling, 0 mode change(s), 0 unsafe hour(s)
	2 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	mechanical	no	dew point	19.035	21.0	18.355	1.481
01	mechanical	no	dew point	18.684	21.0	17.780	1.300
02	mechanical	no	dew point	18.535	21.0	17.781	1.198
03	mechanical	no	dew point	18.641	21.0	17.983	1.194
04	mechanical	yes	minimum dwell	18.446	21.0	17.781	1.093
05	mechanical	yes	minimum dwell	18.733	21.0	17.799	1.231
06	mechanical	no	dew point	19.275	21.0	18.332	1.230

07	mechanical	no	dry-bulb	21.450	21.0	18.893	2.167
08	mechanical	no	dry-bulb	24.006	21.0	20.560	2.732
09	mechanical	no	dry-bulb	24.843	21.0	21.110	2.426
10	mechanical	no	dry-bulb	25.355	21.0	22.223	2.099
11	mechanical	no	dry-bulb	25.912	21.0	23.331	1.620
12	mechanical	no	dry-bulb	26.736	21.0	25.000	1.684
13	mechanical	no	dry-bulb	26.823	21.0	25.560	1.235
14	mechanical	no	dry-bulb	27.482	21.0	26.110	1.355
15	mechanical	no	dry-bulb	27.864	21.0	26.110	1.304
16	mechanical	no	dry-bulb	25.888	21.0	23.330	1.342
17	mechanical	no	dry-bulb	25.629	21.0	23.330	1.454
18	mechanical	no	dry-bulb	24.783	21.0	23.330	1.262
19	mechanical	no	dry-bulb	22.854	21.0	22.780	1.585
20	mechanical	no	dry-bulb	22.288	21.0	21.670	2.003
21	mechanical	no	dry-bulb	22.526	21.0	21.312	2.368
22	mechanical	no	dry-bulb	21.182	21.0	19.440	2.023
23	FREE	yes	-	19.790	21.0	17.780	1.560

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin
actual = what the intake temperature turned out to be, from the station record
margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 19.035 C would have passed. The outside air is too HUMID: the dew-point bound is 16.39 C against a 15.0 C maximum, failing by 1.39 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

01:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.684 C would have passed. The outside air is too HUMID: the dew-point bound is 15.83 C against a 15.0 C maximum, failing by 0.83 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

02:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.535 C would have passed. The outside air is too HUMID: the dew-point bound is 15.83 C against a 15.0 C maximum, failing by 0.83 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

03:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.641 C would have passed. The outside air is too HUMID: the dew-point bound is 15.56 C against a 15.0 C maximum, failing by 0.56 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

04:00 MECHANICAL [minimum dwell]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.446 C against a 21.0 C limit. The schedule is what forbids it: the plant must hold its current mode for 3 h before changing again. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

05:00 MECHANICAL [minimum dwell]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.733 C against a 21.0 C limit. The schedule is what forbids it: the plant must hold its current mode for 3 h before changing again. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

06:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 19.275 C would have passed. The outside air is too HUMID: the dew-point bound is 15.27 C against a 15.0 C maximum, failing by 0.27 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.450 C against a 21.0 C limit, so it fails by 0.450 C. It would take a limit 0.450 C higher, or a bound 0.450 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.006 C against a 21.0 C limit, so it fails by 3.006 C. It would take a limit 3.006 C higher, or a bound 3.006 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.844 C against a 21.0 C limit, so it fails by 3.844 C. It would take a limit 3.844 C higher, or a bound 3.844 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.355 C against a 21.0 C limit, so it fails by 4.355 C. It would take a limit 4.355 C higher, or a bound 4.355 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.911 C against a 21.0 C limit, so it fails by 4.911 C. It would take a limit 4.911 C higher, or a bound 4.911 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.736 C against a 21.0 C limit, so it fails by 5.736 C. It would take a limit 5.736 C higher, or a bound 5.736 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.823 C against a 21.0 C limit, so it fails by 5.823 C. It would take a limit 5.823 C higher, or a bound 5.823 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.482 C against a 21.0 C limit, so it fails by 6.482 C. It would take a limit 6.482 C higher, or a bound 6.482 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 27.864 C against a 21.0 C limit, so it fails by 6.864 C. It would take a limit 6.864 C higher, or a bound 6.864 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.888 C against a 21.0 C limit, so it fails by 4.888 C. It would take a limit 4.888 C higher, or a bound 4.888 C tighter, to change this hour.

17:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.629 C against a 21.0 C limit, so it fails by 4.629 C. It would take a limit 4.629 C higher, or a bound 4.629 C tighter, to change this hour.

18:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.783 C against a 21.0 C limit, so it fails by 3.783 C. It would take a limit 3.783 C higher, or a bound 3.783 C tighter, to change this hour.

19:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.854 C against a 21.0 C limit, so it fails by 1.854 C. It would take a limit 1.854 C higher, or a bound 1.854 C tighter, to change this hour.

20:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.288 C against a 21.0 C limit, so it fails by 1.288 C. It would take a limit 1.288 C higher, or a bound 1.288 C tighter, to change this hour.

21:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.526 C against a 21.0 C limit, so it fails by 1.526 C. It would take a limit 1.526 C higher, or a bound 1.526 C tighter, to change this hour.

22:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.182 C against a 21.0 C limit, so it fails by 0.182 C. It would take a limit 0.182 C higher, or a bound 0.182 C tighter, to change this hour.

23:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 19.790 C, which is 1.210 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.560 C, and the margin is measured, not chosen: 1.494 C of group-conditional forecast error for this hour of day, plus 0.0652 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

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